Rectangular matrix multiplication from polynomial products

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How far can the methods behind the bound $\omega \leq 2.25$ be pushed for rectangular products?

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Based on the accompanying note. Slides prepared with GPT-6.1.

What is ω?

Given $A, B \in \mathbb{C}^{n \times n}$, compute $C = AB$ exactly.

The matrix-multiplication exponent

ω is the infimum of exponents τ for which some algorithm multiplies two $n \times n$ matrices using $O(n^\tau)$ field operations.

The usual algorithm

n^2 dot products of length n:

$$O(n^3), \qquad \omega \leq 3.$$

Strassen's algorithm

Seven recursive products of half-size blocks:

$$O(n^{\log_2 7}), \qquad \omega \leq \log_2 7 \approx 2.80735.$$

There are n^2 outputs, so $\omega \geq 2$. The central open question is whether $\omega = 2$.

We count field operations over $\mathbb{C}$, as in the note. Throughout, $O(n^e)$ suppresses a $+o(1)$ term in the exponent. Strassen: Gaussian elimination is not optimal (1969).

A short history of square matrix multiplication

Year	Algorithm / analysis	Upper bound on ω
Classical	Entry-by-entry dot products	3
1969	Strassen	$\log_2 7 \approx 2.80735$
1990	Coppersmith–Winograd	2.375477
2014	Le Gall	2.3728639
2024	Alman et al.	2.371339
Aug. 2026	Dupont et al.: optimization + AlphaEvolve	2.371177
Sept. 2026	Polynomial-product / completion analysis	2.258
Oct. 2026	Polynomial-product bound [1]	2.25

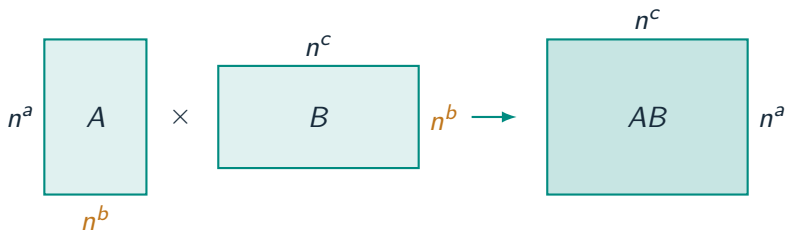
This talk asks how the methods behind the 2.25 bound extend to rectangular products.

What is $\omega(a, b, c)$?

For fixed $a, b, c \geq 0$, consider the product

$$(n^a \times n^b) \cdot (n^b \times n^c) \longrightarrow (n^a \times n^c).$$

Its infimum running-time exponent is $\boldsymbol{\omega(a, b, c)}$.



Exact multiplication over $\mathbb{C}$; dimensions are rounded upward.

The rectangular curve and the dual exponent

Two useful special parameters

$$\omega(k) = \omega(1, k, 1), \qquad \omega(1) = \omega.$$

$$\alpha = \sup\{k \geq 0 : \omega(k) = 2\}.$$

The **dual exponent** α measures how far quadratic-time multiplication reaches as the inner dimension grows.

Rectangular milestones

Year	Source	Bound on α
1997	Coppersmith	> 0.29462
2012	Le Gall	> 0.30298
2018	Le Gall–Urrutia	> 0.31389
2024	Vassilevska Williams et al.	≥ 0.321334
Sept. 2026	OpenAI preprint	> 0.465
This note	**Rectangular extension**	$\mathbf{\geq 0.5}$

Motivation: extend the polynomial-product analysis in [1] to the shapes used by all-pairs algorithms, and make the proof accessible to TCS.

These works also bound the curve $\omega(k)$ beyond the quadratic range; α summarizes where that range ends. Dimensions are rounded upward.

A bound for every matrix shape

Sort the three exponents: $a \geq b \geq c \geq 0$.

Theorem 1.1 of the note

$$\omega(a,b,c) \leq \begin{cases} a+b, & c \leq \dfrac{ab}{a+b}, \\ \dfrac{a+b+c}{2} + \dfrac{1}{4}\left(\dfrac{ab}{c} + \dfrac{ac}{b} + \dfrac{bc}{a}\right), & c > \dfrac{ab}{a+b}. \end{cases}$$

If $c \leq \frac{ab}{a+b}$, the exponent is exactly $a+b$.

Permuting tensor legs lets us use the sorted dimensions of any product.

The largest input size supplies the lower bound $a+b$.

The rectangular curve

Corollary 1.2 of the note

$$\omega(k) \le \begin{cases} 2, & 0 \le k \le \frac{1}{2}, \\ 1 + k + \dfrac{1}{4k}, & k \ge \frac{1}{2}. \end{cases}$$

The thin range is exact.

$\omega(k) = 2$ for $0 \le k \le \frac{1}{2}$.

The dual exponent satisfies $\alpha \ge \frac{1}{2}$.

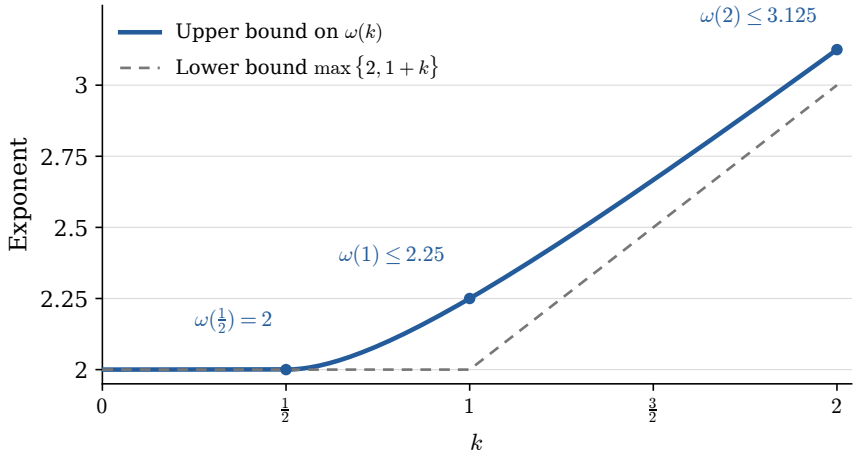
Selected values.

k	1	2
$\omega(k) \le$	2.25	3.125

For $k \ge 1$, the gap above the size lower bound is at most $\frac{1}{4k}$.

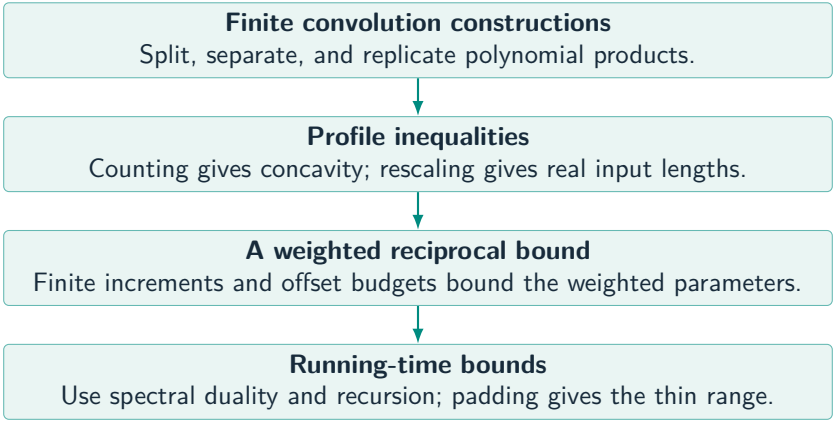
$\alpha = \sup\{k \ge 0 : \omega(k) = 2\}$. The lower bound is $\omega(k) \ge \max\{2, 1 + k\}$.

The bound for $0 \leq k \leq 2$



For $0 \leq k \leq \frac{1}{2}$, the upper bound meets the size lower bound.

The proof in four steps



Sections 2–6 of the note. The spectral theorem is a published input; the rectangular parameter analysis is the extension studied here.

Ordinary block multiplication uses eight products

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}, \qquad C = AB.$$

Four outputs, two products each

$$\begin{aligned} c_{11} &= a_{11}b_{11} + a_{12}b_{21}, \\ c_{12} &= a_{11}b_{12} + a_{12}b_{22}, \\ c_{21} &= a_{21}b_{11} + a_{22}b_{21}, \\ c_{22} &= a_{21}b_{12} + a_{22}b_{22}. \end{aligned}$$

We may use linear combinations

1. Add or subtract blocks within each input.
2. Multiply one combination from each input.
3. Add or subtract the products to form the outputs.

Can linear combinations reduce the eight multiplications?

The entries a_{ij} and b_{ij} may be scalars or matrix blocks. Ordinary block recursion gives $T(n) = 8T(n/2) + O(n^2)$, including all additions.

Strassen's seven products and four outputs

For the same matrices A and B, compute $C = AB$ as follows.

Compute these seven products

$$
\begin{aligned}
M_1 &= (a_{11} + a_{22})(b_{11} + b_{22}),\\
M_2 &= (a_{21} + a_{22})b_{11},\\
M_3 &= a_{11}(b_{12} - b_{22}),\\
M_4 &= a_{22}(b_{21} - b_{11}),\\
M_5 &= (a_{11} + a_{12})b_{22},\\
M_6 &= (a_{21} - a_{11})(b_{11} + b_{12}),\\
M_7 &= (a_{12} - a_{22})(b_{21} + b_{22}).
\end{aligned}
$$

Add and subtract to get the outputs

$$
\begin{aligned}
c_{11} &= M_1 + M_4 - M_5 + M_7,\\
c_{12} &= M_3 + M_5,\\
c_{21} &= M_2 + M_4,\\
c_{22} &= M_1 - M_2 + M_3 + M_6.
\end{aligned}
$$

For example, the first combination simplifies to $c_{11} = a_{11}b_{11} + a_{12}b_{21}$.

Seven block products give $T(n) = 7T(n/2) + O(n^2)$, hence $\omega \leq \log_2 7$.

Strassen (1969): Gaussian elimination is not optimal. Every product has a left-input factor followed by a right-input factor; the same formulas therefore apply to matrix blocks.

Tensor rank counts bilinear products

Use formal labels c_{ij} for the four output positions:

$$T = \underbrace{\begin{aligned} &a_{11}b_{11}c_{11} + a_{12}b_{21}c_{11} + a_{11}b_{12}c_{12} + a_{12}b_{22}c_{12} \\ &+ a_{21}b_{11}c_{21} + a_{22}b_{21}c_{21} + a_{21}b_{12}c_{22} + a_{22}b_{22}c_{22} \end{aligned}}_{\text{8 ordinary terms}}$$

$$\begin{aligned} = {} & (a_{11} + a_{22})(b_{11} + b_{22})(c_{11} + c_{22}) \\ & + (a_{21} + a_{22})b_{11}(c_{21} - c_{22}) + a_{11}(b_{12} - b_{22})(c_{12} + c_{22}) \\ & + a_{22}(b_{21} - b_{11})(c_{11} + c_{21}) + (a_{11} + a_{12})b_{22}(c_{12} - c_{11}) \\ & + (a_{21} - a_{11})(b_{11} + b_{12})c_{22} + (a_{12} - a_{22})(b_{21} + b_{22})c_{11}. \end{aligned}$$

7 Strassen terms: one linear form from each **leg** a, b, c.

Tensor rank is the fewest such terms in a sum for T. This identity gives $\operatorname{rank}(\langle 2,2,2\rangle) \leq 7$.

Section 2. Collect each M_r's coefficients in the four output formulas to obtain its linear form in c. For instance, M_2 multiplies $c_{21} - c_{22}$.

Matrix multiplication tensors and the exponent

All earlier improvements to ω fit this **Strassen-type bilinear framework**, including tensor powers and degenerations.

$$\langle m, r, p \rangle = \sum_{i=1}^{m} \sum_{k=1}^{r} \sum_{j=1}^{p} a_{ik} b_{kj} c_{ij}.$$

It encodes $m \times r$ by $r \times p$ multiplication; c_{ij} labels output position (i, j).

A rank-R identity gives an algorithm using R bilinear products. Applying a square identity with fixed block size q recursively gives

$$T(n) \leq R\, T(n/q) + O(n^2).$$

$$\operatorname{rank}(\langle q, q, q \rangle) \leq R \quad \Longrightarrow \quad \omega \leq \log_q R.$$

Linear-map costs are included; the exponent convention absorbs logarithmic factors.

Restriction: reuse an algorithm after linear substitutions

Make linear substitutions independently on the a-, b-, and c-legs.

What happens to the algorithm?

Linearly encode the new inputs, run the old bilinear products, then linearly combine their outputs.

Each product of three linear forms remains a product of three linear forms after substitution.

Example: keep only the first row

Set $a_{21} = a_{22} = 0$ in 2×2 multiplication:

$$\begin{pmatrix} a_{11} & a_{12} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} \\ 0 & 0 \end{pmatrix}.$$

This computes a 1×2 row times a 2×2 matrix:

$$\langle 2, 2, 2 \rangle \geq \langle 1, 2, 2 \rangle.$$

$$T \geq S \quad \Longrightarrow \quad \mathrm{rank}(S) \leq \mathrm{rank}(T).$$

Section 2. The notation $T \geq S$ means that T restricts to S. Substitutions on the formal output labels specify the linear decoding of outputs. Here the zero second row lets us retain only c_{11}, c_{12}.

Direct sum: run independent computations side by side

Use disjoint variables on all three legs:

$$T(a, b, c) \oplus U(a', b', c') = T(a, b, c) + U(a', b', c').$$

Example: two scalar products

$$a_1 b_1 c_1 + a_2 b_2 c_2$$

encodes the two separate outputs

$$c_1 = a_1 b_1, \qquad c_2 = a_2 b_2.$$

How the algorithms combine

If T uses R products and U uses S products, concatenate the lists: $R + S$ products compute both.

We write mT for m independent copies of T, each with its own inputs and outputs.

$$\operatorname{rank}(T \oplus U) \leq \operatorname{rank}(T) + \operatorname{rank}(U).$$

Why tensor products give block multiplication

Take $q \times q$ blocks, each an $s \times s$ matrix.

Outer	Inner	Paired variable
a_{ik}	a'_{uw}	$a_{(i,u),(k,w)}$
b_{kj}	b'_{wv}	$b_{(k,w),(j,v)}$
c_{ij}	c'_{uv}	$c_{(i,u),(j,v)}$

(i, u) has qs possible values.

The coefficient of $c_{(i,u),(j,v)}$ is

$$\sum_{k=1}^{q} \sum_{w=1}^{s} a_{(i,u),(k,w)}\, b_{(k,w),(j,v)}.$$

This is the usual row–column dot product over all qs inner indices (k, w).

$\langle q, q, q\rangle \otimes \langle s, s, s\rangle \cong \langle qs, qs, qs\rangle$.
Iterating gives $\langle q, q, q\rangle^{\otimes N} \cong \langle q^N, q^N, q^N\rangle$.

Pairing rank-one terms also gives $\operatorname{rank}(T \otimes S) \le \operatorname{rank}(T)\operatorname{rank}(S)$.

Paired variables are independent coordinates. Relabel $(i, u) \leftrightarrow (i-1)s + u$, and similarly for the column and inner indices. Expanding the two tensors gives the displayed output coefficient; q, s, N are positive integers.

Degeneration: a limit of restrictions

Fix tensors T and S: T is the available computation; S is the desired computation.

Definition

A **degeneration** $T \trianglerighteq S$ is a family of linear substitutions on the three legs of T, depending on $\varepsilon \neq 0$, whose resulting tensors $P(\varepsilon)$ converge **coefficient by coefficient** to S as $\varepsilon \to 0$.

$$T \geq P(\varepsilon) \quad (\varepsilon \neq 0), \qquad \lim_{\varepsilon \to 0} P(\varepsilon) = S.$$

For every $a_i b_j c_k$, its coefficient in $P(\varepsilon)$ must tend to its coefficient in S.

Definition: Section 2 of the note. Substitutions act independently on the three legs; all $P(\varepsilon)$ use the same coordinates as S.

A degeneration that cannot be a restriction

Available: $T = a_0' b_0' c_0' + a_1' b_1' c_1'$ (two independent scalar products).

Target: $S = a_0 b_0 c_0 + (a_0 b_1 + a_1 b_0) c_1$.

For $\varepsilon \neq 0$, substitute independently on the three legs:

$$\begin{aligned} a_0' &\mapsto a_0, & b_0' &\mapsto b_0, & c_0' &\mapsto c_0 - c_1/\varepsilon, \\ a_1' &\mapsto a_0 + \varepsilon a_1, & b_1' &\mapsto b_0 + \varepsilon b_1, & c_1' &\mapsto c_1/\varepsilon. \end{aligned}$$

$$\begin{aligned} P(\varepsilon) &= a_0 b_0 \left(c_0 - \frac{c_1}{\varepsilon} \right) + (a_0 + \varepsilon a_1)(b_0 + \varepsilon b_1) \frac{c_1}{\varepsilon} \\ &= S + \varepsilon a_1 b_1 c_1. \end{aligned}$$

The substitutions contain $1/\varepsilon$ and are undefined at zero, but their divergent terms cancel: $P(\varepsilon) \to S$.
$T \trianglerighteq S$, whereas $T \not\geq S$: $\operatorname{rank}(T) = 2 < 3 = \operatorname{rank}(S)$.

All primed variables are independent. The target is the standard W-tensor with its output coordinates exchanged; see [5, proof of Proposition 13, p. 131]. Rank-three proof in backup.

Asymptotic rank is enough to bound ω

The **asymptotic rank** measures the growth of rank under tensor powers:

$$\widetilde{R}(T) := \lim_{N\to\infty} \operatorname{rank}(T^{\otimes N})^{1/N} = \inf_{N\geq 1} \operatorname{rank}(T^{\otimes N})^{1/N}.$$

Fix an integer $q \geq 2$ and put $M = \langle q, q, q\rangle$. A rank decomposition of $M^{\otimes N} \cong \langle q^N, q^N, q^N\rangle$ gives a block algorithm of side q^N. Recursing gives

$$\omega \leq \log_{q^N} \operatorname{rank}(M^{\otimes N}) = \frac{\log_q \operatorname{rank}(M^{\otimes N})}{N}.$$

Taking the infimum over N gives the same conclusion as a rank bound:

$$\widetilde{R}(\langle q, q, q\rangle) \leq R \quad \Longrightarrow \quad \omega \leq \log_q R.$$

Definition: Section 3; algorithm conversion: Section 6. Submultiplicativity of rank ensures the limit exists. Each tensor power and its finite rank identity are fixed before the input-size limit; linear-map costs are included.

Interpolation makes the asymptotic computation exact

Let $P(\varepsilon) = S + \varepsilon P_1 + \cdots + \varepsilon^L P_L$ describe a degeneration $T \trianglerighteq S$. Every nonzero evaluation is a restriction of T.

Take N tensor factors: $P(\varepsilon)^{\otimes N}$ has degree at most NL.

↓

Interpolate at $NL + 1$ nonzero points to recover the constant coefficient $S^{\otimes N}$.

$$\operatorname{rank}(S^{\otimes N}) \leq (NL+1)\operatorname{rank}(T)^N.$$

Take N-th roots: $(NL+1)^{1/N} \to 1$. Thus $\widetilde{R}(S) \leq \operatorname{rank}(T)$.

Lemma 3.1 and Appendix D; standard interpolation conversion [5]. Interpolation is an exact linear combination of evaluations. For matrix multiplication, a fixed tensor power supplies a block algorithm to recurse.

Tensor characters turn constructions into inequalities

Rank and asymptotic rank are subadditive and submultiplicative. Our counting argument uses **exact addition and multiplication**.

A **tensor character** λ assigns nonnegative values with these rules:

Independent computations	$\lambda(S \oplus T) = \lambda(S) + \lambda(T)$
Tensor products	$\lambda(S \otimes T) = \lambda(S)\lambda(T)$
One scalar product	$\lambda(abc) = 1$
Restriction	$T \geq S \implies \lambda(T) \geq \lambda(S)$

Every character gives a lower bound on asymptotic rank: $\lambda(T) \leq \widetilde{R}(T)$.

Section 3; rank properties: [3, Section 4.1]. The rules give $\lambda(T)^N = \lambda(T^{\otimes N}) \leq \mathrm{rank}(T^{\otimes N})$; take N-th roots and the limit.

Strassen's spectral theorem

For every nonzero tensor T over $\mathbb{C}$:

Spectral duality

$$\widetilde{R}(T) = \max_{\lambda} \lambda(T),$$

where the maximum ranges over all tensor characters.

A bound $\lambda(T) \leq R$ for **every** character implies $\widetilde{R}(T) \leq R$.

For $M = \langle q, q, q \rangle$, with integer $q \geq 2$,

$$\lambda(M) \leq R \text{ for every } \lambda \quad \Longrightarrow \quad \omega \leq \log_q R.$$

Theorem 3.2 of the note; Strassen [2, Theorem 3.8] and Wigderson–Zuiddam [3, Corollary 3.30, author version]. The supremum is attained. The exponent implication uses the fixed-power block algorithm from slide 19.

Convolution and dot-product tensors

Write $x^{(X)} = x$, $x^{(Y)} = y$, $x^{(Z)} = z$ for the three groups of variables.

Convolution

$$C_{ij}(r,s) = \sum_{h=0}^{r-1} \sum_{\ell=0}^{s-1} x_h^{(i)} x_\ell^{(j)} x_{h+\ell}^{(k)}.$$

Encodes polynomial multiplication. Inputs have r, s coefficients; the output has $r + s - 1$ coefficients.

Oriented dot products

$$B_X(q) = x \sum_{h=1}^{q} y_h z_h,$$

$$B_Y(q) = y \sum_{h=1}^{q} x_h z_h,$$

$$B_Z(q) = z \sum_{h=1}^{q} x_h y_h.$$

C_{ij}: input legs i, j; output leg k.
B_i: leg i is scalar; the other two legs each have q coordinates.

Definitions: Section 2 of the note; convolution background [4]. The legs i, j, k are distinct elements of $\{X, Y, Z\}$; r, s, q are positive integers.

Three parameters describe dot-product values

Fix an arbitrary tensor character λ. For the dot-product tensor $B_X(q) = x \sum_{j=1}^{q} y_j z_j$, and the other two orientations, pairing indices gives

$$B_i(q) \otimes B_i(r) \cong B_i(qr).$$

Character multiplicativity and monotonicity in q force a power law.

One exponent per orientation determines all dot-product values:

$$\lambda(B_i(q)) = q^{p_i}, \qquad p_i \in [0,1], \quad i \in \{X, Y, Z\}.$$

Section 3. The three parameters need not be equal. Interpolation also gives monotonicity under degeneration. Spectral duality [2, 3] will turn character bounds into rank bounds.

Matrix multiplication factors into three dot products

In each term $a_{ik}b_{kj}c_{ij}$, the a, b, c variables belong to legs X, Y, Z, respectively.

Index	Appears in	Dot-product factor
Row $i \in [A]$	a_{ik}, c_{ij}	$B_Y(A)$
Inner $k \in [B]$	a_{ik}, b_{kj}	$B_Z(B)$
Column $j \in [C]$	b_{kj}, c_{ij}	$B_X(C)$

The subscript names the **scalar leg**. Tensoring pairs the indices:

$$\langle A, B, C\rangle \cong B_Y(A) \otimes B_Z(B) \otimes B_X(C).$$

$$\lambda(\langle A, B, C\rangle) = \lambda(B_Y(A))\lambda(B_Z(B))\lambda(B_X(C)) = A^{p_Y} B^{p_Z} C^{p_X}.$$

Sections 2–3. Pairing the factor coordinates gives precisely a_{ik}, b_{kj}, c_{ij}; coordinates on scalar legs have dimension one. The last line uses character multiplicativity and $\lambda(B_i(q)) = q^{p_i}$. Dimensions A, B, C are positive integers.

Shared sectors give a power-sum inequality

The pieces may share variables on leg i; their variables on the other two legs belong to disjoint, matched sectors.

Example: a shared first leg

$$T = \underbrace{xy_1z_1}_{T_1} + \underbrace{xy_2z_2}_{T_2}.$$

This is $B_X(2)$, so

$$\lambda(T) = 2^{p_X},$$
$$\lambda(T_1) = \lambda(T_2) = 1.$$

The general rule, for $p_i > 0$

If $T \trianglerighteq \sum_h T_h$ has this sector structure,

$$\lambda(T)^{1/p_i} \geq \sum_h \lambda(T_h)^{1/p_i}.$$

The next slides prove this using tensor powers, finite separation, and counting.

Lemma 3.3. The common variable x makes the example a shared-leg sum. The following slides prove the rule using finite separation and multinomial counting. The separation construction is in the holdout slides.

Tensor powers group sector choices by multiplicity

Suppose $T \trianglerighteq \sum_{h=1}^{s} T_h$ has matched sectors sharing leg i. Fix a character λ with $p = p_i > 0$.

Expanding $(\sum_h T_h)^{\otimes N}$ gives a sector for every word $w_1 \cdots w_N \in \{1, \ldots, s\}^N$. For example, 112 means $T_1 \otimes T_1 \otimes T_2$:

$$\underbrace{112, \quad 121, \quad 211}_{\text{multiplicities } (n_1,n_2)=(2,1)} \; .$$

Fixing counts $n_1 + \cdots + n_s = N$ gives

$$M = \binom{N}{n_1, \ldots, n_s}, \qquad U = \bigotimes_{h=1}^{s} T_h^{\otimes n_h}.$$

Retain these M words by zeroing coordinates on the two unshared legs. Each word has the factors of U, in a different order.

Section 3, proof of Lemma 3.3. Here $N \geq 1$ and the $n_h \geq 0$ are integers. Independent reordering of the words' factors becomes valid after applying the separation lemma.

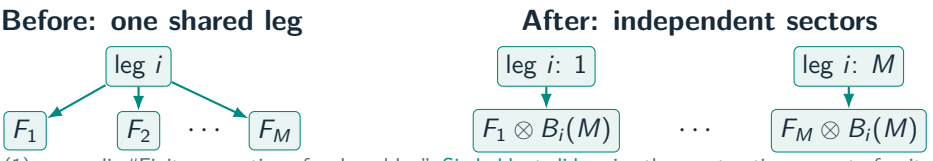
Finite separation: give every sector its own variables

Let $Q = F_1 + \cdots + F_M$, with $M \geq 1$ sectors sharing leg i. Their variables on each of the other two legs are pairwise disjoint.

Finite-separation lemma

$$(2M)Q \trianglerighteq \bigoplus_{w=1}^{M} (F_w \otimes B_i(M)).$$

Here $(2M)Q$ means $2M$ **independent copies** of Q.



Lemma 2.1(1); appendix "Finite separation of a shared leg". Six holdout slides give the construction: a root-of-unity filter of length $2M$, followed by ε-weights that retain matching sector labels.

Finite separation makes the words independent

Let $Q = \sum_{w=1}^{M} F_w$ be the retained word sectors. The previous retention step gives $T^{\otimes N} \trianglerighteq Q$; these sectors share leg i.

Apply the lemma, then reorder factors

$$(2M)\,T^{\otimes N} \trianglerighteq (2M)\,Q \trianglerighteq \bigoplus_{w=1}^{M} (F_w \otimes B_i(M)) \cong \bigoplus_{w=1}^{M} (U \otimes B_i(M)).$$

All three legs now distinguish w, so each word's factors can be reordered independently to give U.

First step: retain the multiplicity group in each copy. Second: the finite-separation lemma (Lemma 2.1(1)), proved in six holdout slides. Third: all words have the same factor multiplicities, so $F_w \cong U$.

A multiplicity group gives one character inequality

Apply λ to the separation degeneration, using monotonicity from interpolation. Since $p = p_i > 0$, we have $\lambda(B_i(M)) = M^p$:

$$2M\,\lambda(T)^N \geq \underbrace{M}_{\text{copies}} \underbrace{\prod_h \lambda(T_h)^{n_h}}_{\lambda(U)} \underbrace{M^p}_{\lambda(B_i(M))}.$$

Cancel M, take the power $1/p$, and set

$$a_h = \lambda(T_h)^{1/p}.$$

$$M \prod_{h=1}^{s} a_h^{n_h} \leq 2^{1/p} \lambda(T)^{N/p}.$$

The bound on the right is the same for every multiplicity tuple.

Lemma 3.3 and the appendix "Counting and rescaling". The calculation uses character additivity, multiplicativity, and monotonicity under degeneration; the latter follows from interpolation (Lemma 3.1; also shown in backup).

Sum over multiplicities, then take roots

There are at most $(N+1)^s$ tuples with $n_1 + \cdots + n_s = N$. Sum their bounds and use the multinomial theorem:

$$\left(\sum_{h=1}^{s} a_h\right)^N = \sum_{\substack{n_1+\cdots+n_s=N \\ n_h \geq 0}} \binom{N}{n_1, \ldots, n_s} \prod_h a_h^{n_h}$$
$$\leq (N+1)^s 2^{1/p} \lambda(T)^{N/p}.$$

Take N-th roots and let $N \to \infty$:

$$\sum_h a_h \leq \underbrace{(N+1)^{s/N} 2^{1/(pN)}}_{\longrightarrow 1} \lambda(T)^{1/p}.$$

This proves Lemma 3.3: $\lambda(T)^{1/p_i} \geq \sum_h \lambda(T_h)^{1/p_i}$.

Lemma 3.3 and the appendix "Counting and rescaling". The number of sectors s and the character's parameter $p = p_i > 0$ stay fixed as $N \to \infty$.

Two-sector splitting lemma

Fix integers $r \geq 1$, $s \geq 2$, and distinct legs i, j.

Two matched sectors

$$C_{ij}(r,s) \otimes B_i(2) \trianglerighteq T_+ + T_-, \qquad T_\pm \cong C_{ij}(r, s \pm 1).$$

The sectors share leg i and use disjoint coordinates on the other legs.

Lemma 2.1(2); [1, Lemma 4.1]. Coordinate interpretation and proof in backup.

Three-sector splitting lemma

Fix integers $r, s \geq 1$ and distinct legs i, j, k.

Three matched sectors

$$C_{ij}(r, 3s + r - 1) \trianglerighteq T_1 + T_2 + T_3,$$
$$T_1, T_2 \cong C_{ij}(r, s), \qquad T_3 \cong C_{ik}(r, s).$$

The sectors share leg i and use pairwise disjoint coordinates on the other legs.

Lemma 2.1(3); [1, Lemma 4.2]. Coordinate interpretation and proof in backup.

The splittings give two character inequalities

Fix a character λ, put $p = p_i > 0$, and write $f_{ij}(r,s) := \lambda(C_{ij}(r,s))$.

Two sectors: neighboring lengths

Since $\lambda(B_i(2)) = 2^p$,

$$2f_{ij}(r,s)^{1/p} \geq f_{ij}(r,s+1)^{1/p} + f_{ij}(r,s-1)^{1/p}.$$

This is discrete concavity of $s \mapsto f_{ij}(r,s)^{1/p}$.

Three sectors: coupled orientations

$$f_{ij}(r,3s+r-1)^{1/p} \geq 2f_{ij}(r,s)^{1/p} + f_{ik}(r,s)^{1/p}.$$

The same parameter p_i controls both orientations, since leg i is shared.

Lemmas 2.1 and 3.3. Character multiplicativity evaluates the two-sector source; the power-sum rule evaluates the shared sectors. The first inequality requires $r \geq 1, s \geq 2$; the second $r, s \geq 1$.

Replication lets us compare all scales

Replication into m independent copies gives

$$C_{ij}(mr, ms) \trianglerighteq mC_{ij}(r, s) \quad \Longrightarrow \quad f_{ij}(mr, ms) \geq mf_{ij}(r, s).$$

Normalize the lengths; the limit exists for $r, s > 0$:

$$g_{ij}(r, s) = \lim_{m \to \infty} \frac{f_{ij}(\lceil mr \rceil, \lceil ms \rceil)}{m}.$$

For $p = p_i > 0$, powered concavity ($0 < \delta < s$) and tripling become

$$2g_{ij}(r, s)^{1/p} \geq g_{ij}(r, s - \delta)^{1/p} + g_{ij}(r, s + \delta)^{1/p},$$
$$g_{ij}(r, 3s + r)^{1/p} \geq 2g_{ij}(r, s)^{1/p} + g_{ik}(r, s)^{1/p}.$$

The additive -1 becomes $-1/m \to 0$ after rescaling.

The profiles are continuous and homogeneous: $g_{ij}(tr, ts) = t\, g_{ij}(r, s)$.

Lemmas 2.2 and 4.1; Appendices E–F. Polynomial division and evaluation give replication. Common-scale approximation preserves the inequalities; additional profile properties are in backup.

Concavity and tripling couple two profiles

Fix a leg i with $p_i > 0$; let j, k be the other legs. Write

$$P(s) := P_{ij}(s) = g_{ij}(1, s)^{1/p_i}, \qquad Q(s) := P_{ik}(s) = g_{ik}(1, s)^{1/p_i}.$$

Both profiles are positive and nondecreasing for real $s > 0$. Their inequalities are

Concavity, $0 < \delta < s$

$$2P(s) \geq P(s - \delta) + P(s + \delta),$$
$$2Q(s) \geq Q(s - \delta) + Q(s + \delta).$$

Tripling, $s > 0$

$$P(3s + 1) \geq 2P(s) + Q(s),$$
$$Q(3s + 1) \geq 2Q(s) + P(s).$$

We will prove that $\lim_{s\to\infty} \frac{P(s)}{s} = \lim_{s\to\infty} \frac{Q(s)}{s} = L_i > 0$.

Rescaled-profile inequalities (Lemma 4.1), with $r = 1$. The second tripling inequality exchanges j and k; the parameter p_i stays the same.

Tripling forces the limiting slopes to agree

The secant slopes $(P(s) - P(1))/(s-1)$ are nonnegative and decrease with $s > 1$. Thus their finite limit exists and equals

$$L_P := \lim_{s\to\infty} \frac{P(s) - P(1)}{s-1} = \lim_{s\to\infty} \frac{P(s)}{s} \geq 0.$$

Define L_Q analogously for Q.

Divide both tripling inequalities by s and let $s \to \infty$. Since $P(3s+1)/s \to 3L_P$, and likewise for Q,

$$3L_P \geq 2L_P + L_Q \quad \Longrightarrow \quad L_P \geq L_Q,$$
$$3L_Q \geq 2L_Q + L_P \quad \Longrightarrow \quad L_Q \geq L_P.$$

$L_P = L_Q =: L_i \geq 0$. It remains to prove $L_i > 0$.

Proof of Lemma 5.1, Appendix G. Monotonicity makes the secants nonnegative; concavity makes them decrease.

Concavity gives the sum bound directly

Put $R = P + Q > 0$. For $0 < u < S$, concavity and tripling give

$$\frac{P(S) - P(u)}{S - u} \geq \frac{P(3S+1) - P(S)}{2S+1} \geq \frac{R(S)}{2S+1}.$$

The same bound holds for Q. Adding and rearranging,

$$R(S) - R(u) \geq \frac{2(S-u)R(S)}{2S+1} \quad \Longrightarrow \quad \frac{R(u)}{2u+1} \leq \frac{R(S)}{2S+1}.$$

Send $S \to \infty$, using $R(S)/(2S+1) \to L_i$:

$$R(u) \leq L_i(2u+1), \quad \text{hence } L_i \geq \frac{R(u)}{2u+1} > 0.$$

Proof of Lemma 5.1, Appendix G. Concavity compares the secant on $[u, S]$ with the later secant on $[S, 3S+1]$. The sum bound will also control the offsets.

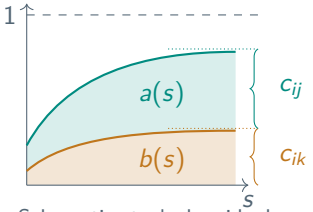
The two residuals share a budget of one

Divide $P(s) + Q(s) \le L_i(2s+1)$ by $L_i > 0$ and subtract $2s$:

$$\underbrace{\left(\frac{P(s)}{L_i} - s\right)}_{a(s)} + \underbrace{\left(\frac{Q(s)}{L_i} - s\right)}_{b(s)} \le 1.$$

Concavity makes a, b nonnegative and nondecreasing. The budget bounds each by 1, so both limits exist:

$$c_{ij} := \lim_{s\to\infty} a(s),$$
$$c_{ik} := \lim_{s\to\infty} b(s).$$



Schematic stacked residuals

$c_{ij}, c_{ik} \ge 0, \quad c_{ij} + c_{ik} \le 1.$
$L_i s \le P(s) \le L_i(s + c_{ij}), \quad L_i s \le Q(s) \le L_i(s + c_{ik}).$

Lemma 5.1, Appendix G. The limits exist because the residuals are nondecreasing and bounded; $a(s) \le c_{ij}$ and $b(s) \le c_{ik}$.

Finite increments give relative growth

Assume $p_i > 0$. The common-slope lemma gives, for $s, t > 0$,

$$P_{ij}(s+t) \geq P_{ij}(s) + L_i t, \qquad P_{ij}(s) \leq L_i(s + c_{ij}).$$

Divide the first inequality by $P_{ij}(s) > 0$ and use the second:

$$\frac{P_{ij}(s+t)}{P_{ij}(s)} \geq 1 + \frac{L_i t}{P_{ij}(s)} \geq 1 + \frac{t}{s + c_{ij}}.$$

Since $g_{ij}(r, s) = r P_{ij}(s/r)^{p_i}$, homogeneity gives

$$\frac{g_{ij}(r, s+t)}{g_{ij}(r, s)} \geq \left(1 + \frac{t}{s + c_{ij} r}\right)^{p_i} \qquad (r, s, t > 0).$$

Lemma 5.1 and Proposition 5.2. The lower bound applies to every positive finite increment.

Homogeneity gives the pair inequality

Assume $p_i, p_j > 0$. Increase r by εr, then s by εs. Input interchange gives the first increment bound:

$$1+\varepsilon = \frac{g_{ij}((1+\varepsilon)r, (1+\varepsilon)s)}{g_{ij}(r,s)} \geq \left(1+\frac{\varepsilon r}{r+c_{ji}s}\right)^{p_j} \left(1+\frac{\varepsilon s}{s+c_{ij}(1+\varepsilon)r}\right)^{p_i}.$$

Take logarithms, divide by ε, and let $\varepsilon \downarrow 0$:

$$\frac{p_i s}{s+c_{ij}r} + \frac{p_j r}{r+c_{ji}s} \leq 1 \qquad (r,s>0).$$

Proposition 5.2. The only limit used is $\log(1+\varepsilon x)/\varepsilon \to x$. For $p_i = 0$, set its two offsets to zero; pairs involving it satisfy the bound directly because the remaining summand is at most $p_j \leq 1$.

The weighted reciprocal bound

For any positive weights w_1, w_2, w_3, our target is

$$\sum_i w_i p_i \le \frac{1}{4}\left(\sum_i \frac{1}{w_i}\right)\left(\sum_{i<j} w_i w_j\right).$$

For each unordered pair $\{i, j\}$, set $r = 1/w_i$, $s = 1/w_j$ in the pair inequality and multiply by $w_i w_j$:

$$w_i w_j \ge \frac{p_i w_i^2}{w_i/w_j + c_{ij}} + \frac{p_j w_j^2}{w_j/w_i + c_{ji}}.$$

We will sum the three inequalities and use $c_{ij} + c_{ik} \le 1$.

Proposition 5.3. The offsets are fixed independently of the chosen lengths and weights. No equality among the parameters is assumed.

The reciprocal inequality eliminates the offsets

Summing over the three unordered pairs and grouping by p_i gives

$$\sum_{i<j} w_i w_j \geq \sum_i p_i w_i^2 \left(\frac{1}{w_i/w_j + c_{ij}} + \frac{1}{w_i/w_k + c_{ik}} \right).$$

Here j, k are the other indices in the summand indexed by i. For positive x, y,

$$\frac{1}{x} + \frac{1}{y} \geq \frac{4}{x+y}.$$

Apply this and $c_{ij} + c_{ik} \leq 1$:

$$\sum_{i<j} w_i w_j \geq 4 \sum_i \frac{p_i w_i^2}{w_i/w_j + w_i/w_k + 1} = \frac{4 \sum_i w_i p_i}{\sum_{\ell=1}^{3} 1/w_\ell}.$$

This proves the weighted bound.

Proposition 5.3. All denominators are positive, and $p_i \geq 0$ preserves each lower bound.

Character bounds give arithmetic algorithms

Published spectral input:

$$\widetilde{R}(T) = \max_{\lambda} \lambda(T), \qquad \widetilde{R}(T) = \inf_{N \geq 1} \operatorname{rank}(T^{\otimes N})^{1/N}.$$

If every character has $ap_Y + bp_Z + cp_X \leq E$:

1. Choose a fixed large base B and round the block dimensions upward.
2. Spectral duality supplies a finite exact rank identity for a tensor power.
3. Apply that identity recursively to matrix blocks; pad to the input size.

For every positive exponent slack, we obtain a fixed finite block algorithm. Recursion gives the claimed arithmetic exponent.

Theorem 3.2 [2, 3] and Lemma 6.1. Linear combinations cost $O(\ell R^{\ell})$ over ℓ recursion levels. The exponent statement does not provide a practical small-base implementation.

Substitute the shape into the weighted bound

For $a, b, c > 0$, take $w = (a, b, c)$ and $(p_1, p_2, p_3) = (p_Y, p_Z, p_X)$. For every character,

$$\begin{aligned} ap_1 + bp_2 + cp_3 &\le \frac{1}{4}\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)(ab + ac + bc) \\ &= \frac{a+b+c}{2} + \frac{1}{4}\left(\frac{ab}{c} + \frac{ac}{b} + \frac{bc}{a}\right). \end{aligned}$$

The character-to-algorithm implication now gives

$$\omega(a, b, c) \le \frac{a+b+c}{2} + \frac{1}{4}\left(\frac{ab}{c} + \frac{ac}{b} + \frac{bc}{a}\right).$$

Proposition 5.3 and Lemma 6.1. Zero character parameters were included in the pair inequality; this bound holds for every positive shape, including the thin threshold.

The thin case follows by padding

Assume $a \geq b > 0$ and put

$$c_0 = \frac{ab}{a+b}.$$

Padding makes ω nondecreasing in each argument. For $0 \leq c \leq c_0$,

$$\omega(a,b,c) \leq \omega(a,b,c_0) \leq a+b.$$

Proof of Theorem 1.1. The bound on the preceding slide applies at $c_0 > 0$. The matching size lower bound gives $\omega(a,b,c) = a+b$ throughout the thin range.

Applications through established tradeoffs

Problem	Running time
Directed unweighted APSP: Zwick's balance [6]	$O(n^{2.5})$
Combining with Alman–Vassilevska Williams [7]	$O(n^{2.4999})$
One LCA per pair in a DAG [8]	$O(n^{(3+\sqrt{3})/2})$
Exact all-pairs Hamming distance [9, 10]	$O(n^{(13-\sqrt{7})/4})$
Minimum / maximum Boolean witnesses [11]	$O(n^{2.5})$
Sparse multiplication, $m \leq n^{1.25}$ [12]	$O(n^2)$

Section 7. The LCA and Hamming exponents are approximately 2.36603 and 2.58856, respectively. Hamming inputs are n vectors of length n over an unrestricted alphabet; sparse inputs have at most m nonzeros each.

What the rectangular extension contributes

Starting constructions

The two polynomial splittings recap [1]. Finite separation adapts its shared-leg construction.

Standard tools

Multinomial counting, spectral duality, interpolation, the reciprocal inequality, and padding.

Rectangular analysis

Keep all orientations; replicate and rescale convolution profiles; couple six offsets; sum the pair inequalities at reciprocal lengths.

Lean correspondence

The main bounds are checked with the published spectral theorem supplied explicitly as a hypothesis.

$$c \le \frac{ab}{a+b} \implies \omega(a,b,c) = a+b \qquad (a \ge b \ge c \ge 0).$$

The note makes no priority claim for standard tools or finite construction variants. Lean checks mathematical statements; some finite proof routes differ from the manuscript. The application algorithms are outside the formalization.

Backup: the degeneration target has rank three

The coefficients of c_0 and c_1 in $S = a_0 b_0 c_0 + (a_0 b_1 + a_1 b_0) c_1$ are the two matrices

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

A rank-two decomposition of S would put both A and B in the span of two rank-one matrices. Since A, B are independent, those matrices would have to span exactly the same plane.

$$\alpha A + \beta B = \begin{pmatrix} \alpha & \beta \\ \beta & 0 \end{pmatrix}, \qquad \det(\alpha A + \beta B) = -\beta^2.$$

Every nonzero rank-one matrix in this plane is therefore a multiple of A. Two independent rank-one generators cannot exist.

Thus $\operatorname{rank}(S) > 2$. Its three displayed monomials give $\operatorname{rank}(S) \leq 3$, so $\operatorname{rank}(S) = 3$.

Elementary slice-space proof over $\mathbb{C}$; the same rank statement is in [5, proof of Proposition 13, p. 131]. In a rank-two decomposition, the two bilinear factors are rank-one matrices; both output coefficients are linear combinations of them. Return to the example.

Separation proof: start with a shared first leg

Work over $\mathbb{C}$. Assume leg X is shared, and write

$$Q = \sum_{h=1}^{M} F_h, \qquad F_h = \sum_{a,b,c} \mu_{habc}\, x_a y_{h,b} z_{h,c}.$$

The x_a may be shared across sectors. The $y_{h,b}$ and $z_{h,c}$ are already private to sector h.

The target has independent variables on every leg

$$S := \bigoplus_{h=1}^{M} (F_h \otimes B_X(M)).$$

We construct an exact restriction, then a degeneration:

$$(2M)Q \;\geq\; R \;\trianglerighteq\; S.$$

Proof of Lemma 2.1(1), appendix "Finite separation of a shared leg". Here $M \geq 1$; all coordinate sets are finite. Other shared-leg orientations follow by permuting the three legs.

Mix $2M$ copies using roots of unity

Set $L = 2M$, and choose a primitive L-th root of unity ζ. Label the independent copies by $\ell = 0, \ldots, L-1$.

In copy ℓ, make these linear substitutions:

$$x_a^{(\ell)} \longmapsto \sum_{g=1}^{M} \zeta^{\ell g} X_{a,g},$$

$$y_{h,b}^{(\ell)} \longmapsto \sum_{u=1}^{M} \zeta^{\ell(u-h)} Y_{h,b,u},$$

$$z_{h,c}^{(\ell)} \longmapsto \frac{1}{L} \sum_{v=1}^{M} \zeta^{-\ell v} Z_{h,c,v}.$$

The new indices g, u, v each run from 1 to M. Each substitution acts on just one leg.

All new coordinates are independent within each leg. The first-leg label is g; u, v label coordinates on the other legs. Summing copies cancels unwanted combinations of these labels.

The finite sum keeps exactly $g + u = h + v$

After summing the copies, the coefficient of $X_{a,g}Y_{h,b,u}Z_{h,c,v}$ is μ_{habc} times

$$\frac{1}{L}\sum_{\ell=0}^{L-1}\zeta^{\ell(g+u-h-v)} = \begin{cases} 1, & g+u-h-v \equiv 0 \pmod{L}, \\ 0, & \text{otherwise.} \end{cases}$$

Since $|g + u - h - v| \leq 2M - 2 < L$, the only multiple of L in this range is zero. Thus the filter keeps precisely $g + u = h + v$.

The exact restriction obtained so far

$$(2M)Q \geq R := \sum_{\substack{h,g,u,v \\ g+u=h+v}} \sum_{a,b,c} \mu_{habc} X_{a,g} Y_{h,b,u} Z_{h,c,v}.$$

The filter identity is a finite geometric sum: if $\zeta^d \neq 1$, then $\sum_{\ell=0}^{L-1}(\zeta^d)^\ell = 0$; otherwise the sum is L. All h, g, u, v lie in $\{1, \ldots, M\}$.

Choose weights whose sum is a square

Every monomial of R satisfies $g + u = h + v$. For $\varepsilon \neq 0$, scale its coordinates by

$$\begin{aligned} X_{a,g} &\longmapsto \varepsilon^{g^2} X_{a,g}, \\ Y_{h,b,u} &\longmapsto \varepsilon^{2hu-h^2} Y_{h,b,u}, \\ Z_{h,c,v} &\longmapsto \varepsilon^{-2hv} Z_{h,c,v}. \end{aligned}$$

A surviving monomial acquires the total exponent

$$\begin{aligned} g^2 + 2hu - h^2 - 2hv &= g^2 + 2h(u - v) - h^2 \\ &= g^2 + 2h(h - g) - h^2 = (g - h)^2 \geq 0. \end{aligned}$$

The exponent is zero exactly when $g = h$; all other retained monomials have positive exponent.

Each weight depends only on indices of the coordinate being scaled, so the three substitutions act independently on the legs. The relation $u - v = h - g$ comes from the preceding filter.

Take the limit: matching labels survive

After scaling, the resulting tensor is

$$P(\varepsilon) = \sum_{\substack{h,g,u,v \\ g+u=h+v}} \sum_{a,b,c} \mu_{habc}\, \varepsilon^{(g-h)^2} X_{a,g}\, Y_{h,b,u} Z_{h,c,v}.$$

As $\varepsilon \to 0$, only $g = h$ survives. The filter equation then forces $u = v$:

$$\lim_{\varepsilon \to 0} P(\varepsilon) = \sum_{h=1}^{M} \sum_{a,b,c} \sum_{u=1}^{M} \mu_{habc} X_{a,h} Y_{h,b,u} Z_{h,c,u} =: S.$$

For $\varepsilon \neq 0$, $R \geq P(\varepsilon)$. The coefficientwise limit proves $R \trianglerighteq S$, hence $(2M)Q \trianglerighteq S$.

Negative powers are allowed in the substitutions for nonzero ε. After the finite filter, every nonzero coefficient has the nonnegative exponent $(g-h)^2$, so the limit exists.

Identify the independent sectors and dot products

Write $S = \sum_{h=1}^{M} S_h$, where

$$S_h = \sum_{a,b,c} \sum_{u=1}^{M} \mu_{habc} X_{a,h} Y_{h,b,u} Z_{h,c,u}.$$

Recall $B_X(M) = s \sum_{u=1}^{M} t_u w_u$. For each fixed h, pair the coordinates in $F_h \otimes B_X(M)$:

F_h coordinate	$B_X(M)$ coordinate	Paired coordinate
x_a	s	$X_{a,h}$
$y_{h,b}$	t_u	$Y_{h,b,u}$
$z_{h,c}$	w_u	$Z_{h,c,u}$

Thus $S_h \cong F_h \otimes B_X(M)$. All three legs carry h, so different S_h use disjoint variables.

$$(2M)Q \trianglerighteq S \cong \bigoplus_{h=1}^{M} (F_h \otimes B_X(M)).$$

This completes the proof of Lemma 2.1(1). Permuting the legs gives the same construction for any shared leg i.

Two-sector proof: the source and target maps

Fix $r \geq 1, s \geq 2$. Let $\mathcal{P}_d = \{p(t) : \deg p < d\}$ and $f \in \mathcal{P}_r$.

Source: lengths s, s

$$(f, (a, b)) \longmapsto (fa, fb),$$
$$(a, b) \in \mathcal{P}_s \oplus \mathcal{P}_s.$$

Target: lengths $s+1, s-1$

$$(f, (p, h)) \longmapsto (fp, fh),$$
$$(p, h) \in \mathcal{P}_{s+1} \oplus \mathcal{P}_{s-1}.$$

The first input f is shared. The second input contains two independent polynomials; the two output blocks are disjoint.

Lemma 2.1(2), appendix "Splitting two polynomial products". Return to the lemma statement.

Why tensoring with $B_X(2)$ gives two products

Recall: $B_X(2) = (uv_0)w_0 + (uv_1)w_1$ encodes outputs (uv_0, uv_1).

Read X, Y as inputs and Z as output. Both factors are bilinear maps:

$$C_{XY}(r,s) : \mathcal{P}_r \times \mathcal{P}_s \to \mathcal{P}_{r+s-1}, \qquad (f,g) \longmapsto fg,$$
$$B_X(2) : \mathbb{C} \times \mathbb{C}^2 \to \mathbb{C}^2, \qquad (u,(v_0,v_1)) \longmapsto (uv_0, uv_1).$$

The tensor product pairs corresponding input and output spaces:

$$\underbrace{(\mathcal{P}_r \otimes \mathbb{C})}_{\mathcal{P}_r} \times \underbrace{(\mathcal{P}_s \otimes \mathbb{C}^2)}_{\mathcal{P}_s \oplus \mathcal{P}_s} \longrightarrow \underbrace{(\mathcal{P}_{r+s-1} \otimes \mathbb{C}^2)}_{\mathcal{P}_{r+s-1} \oplus \mathcal{P}_{r+s-1}}.$$

For the standard basis e_0, e_1 of $\mathbb{C}^2$, this map sends

$$(f \otimes 1,\ a \otimes e_0 + b \otimes e_1) \longmapsto fa \otimes e_0 + fb \otimes e_1.$$

Thus $(f,(a,b)) \mapsto (fa, fb)$: two products sharing f.

The parameters $r, s, 2$ specify dimensions, not function arguments.

Two-sector proof: encode the inputs

Fix arbitrary $r \geq 1, s \geq 2$. The inputs are $f \in \mathcal{P}_r$ and $(p, h) \in \mathcal{P}_{s+1} \oplus \mathcal{P}_{s-1}$. Keep f unchanged.

Write $p_s = [t^s]p$. For $\varepsilon \neq 0$, encode the second input by

$$E_\varepsilon : (p, h) \longmapsto (a, b), \qquad \begin{cases} a = p - p_s t^s - \varepsilon^{-1} th, \\ b = \varepsilon^{-1} h + p_s t^{s-1}. \end{cases}$$

Both have degree at most $s - 1$: the leading term of p was removed, and $\deg(th) \leq s - 1$.

Apply the source bilinear map:

$$(f, (p, h)) \xrightarrow{\mathrm{id} \times E_\varepsilon} (f, (a, b)) \longmapsto (U, V) = (fa, fb).$$

Lemma 2.1(2), appendix "Splitting two polynomial products". E_ε is a linear map from $\mathcal{P}_{s+1} \oplus \mathcal{P}_{s-1}$ to $\mathcal{P}_s \oplus \mathcal{P}_s$, independently of f. The definitions give $a + tb = p$.

Two-sector proof: decode the outputs

Put $m = r + s - 2$, so $\deg U, \deg V \le m$. Decode the output pair linearly:

$$D_\varepsilon : (U, V) \longmapsto (P, H_\varepsilon), \qquad \begin{cases} P = U + tV, \\ H_\varepsilon = \varepsilon(V - [t^m]V\, t^m). \end{cases}$$

The output lengths are $m + 2 = r + s$ for P and $m = r + s - 2$ for H_ε.

Since $a + tb = p$,

$$P = U + tV = f(a + tb) = fp.$$

Expand the other product:

$$V = \varepsilon^{-1} fh + p_s t^{s-1} f.$$

Because $\deg(fh) \le r + s - 3 = m - 1$, its top coefficient is

$$[t^m]V = p_s f_{r-1}, \qquad f_{r-1} = [t^{r-1}]f.$$

Coefficient extraction, deleting a coefficient, and shifting polynomial coefficients are linear output operations. The identity $[t^m]V = p_s f_{r-1}$ is used to analyze that output map.

Two-sector proof: the error vanishes

Substitute the expression for V and its top coefficient:

$$\begin{aligned} H_\varepsilon &= \varepsilon\big(V - p_s f_{r-1} t^{r+s-2}\big) \\ &= fh + \varepsilon p_s t^{s-1}\big(f - f_{r-1} t^{r-1}\big) \xrightarrow{\varepsilon \to 0} fh. \end{aligned}$$

Encoding the second input, applying the source map, and decoding gives

$$(f, (p, h)) \longmapsto (fp,\ \ fh + \varepsilon p_s t^{s-1}(f - f_{r-1} t^{r-1})).$$

The coefficients converge to those of (fp, fh). The independent inputs p, h have lengths $s+1, s-1$, and the two output blocks are disjoint; both products share f.

$$C_{XY}(r,s) \otimes B_X(2) \trianglerighteq T_+ + T_-, \qquad T_\pm \cong C_{XY}(r, s \pm 1).$$

This proves Lemma 2.1(2) for arbitrary $r \geq 1, s \geq 2$. If $r = 1$, the error is identically zero. Permuting the legs gives every orientation i, j.

Three-sector proof: inputs and outputs

Fix $r, s \geq 1$ and use $i = X, j = Y, k = Z$. Write $\mathcal{P}_d = \{p(t) : \deg p < d\}$.

The source map is $(f, g) \mapsto fg$, with $f \in \mathcal{P}_r$ and $g \in \mathcal{P}_{3s+r-1}$.

The sectors share f; their Y, Z coordinate blocks are disjoint:

Sector	Leg j	Leg k
T_1, T_2	Input: s coordinates	Output: $r + s - 1$ coordinates
T_3	Output: $r + s - 1$ coordinates	Input: s coordinates

Lemma 2.1(3), appendix "The three-sector splitting". The middle sector's first-leg reversal identifies that individual tensor. Return to the lemma statement.

Three sectors, with X, Y kept as the input legs

Split the second input into independent coefficient blocks:

$$g = g_L + t^s q + t^{2s+r-1} g_R, \qquad g_L, g_R \in \mathcal{P}_s, \quad q \in \mathcal{P}_{r+s-1}.$$

The target of the degeneration is the bilinear map

$$(f, (g_L, q, g_R)) \longmapsto \underbrace{(fg_L,\ M_f(q),\ fg_R)}_{\text{three output blocks}}, \qquad f \in \mathcal{P}_r,$$

where the middle output keeps s coefficients of fq:

$$M_f(q) := \sum_{v=0}^{s-1} [t^{r-1+v}](fq)\, t^v \in \mathcal{P}_s.$$

The outer outputs fg_L, fg_R each lie in $\mathcal{P}_{r+s-1}$.

There are two input groups: f and (g_L, q, g_R). The name C_{XZ} for the middle sector reads Z as input and Y as output; the proof below makes this identification explicit.

Here $[t^k]$ extracts the coefficient of t^k. The same three-leg tensor can be read with different choices of output leg.

Three-sector proof: partition the coordinates

Let x_a, y_b be coefficients of f, g, and z_c output labels. Set $n = 3s + r - 1$:

$$C_{XY}(r, n) = \sum_{a=0}^{r-1} \sum_{b=0}^{n-1} x_a y_b z_{a+b}.$$

Partition the integer indices of Y and Z into these disjoint intervals:

	Left	Middle	Right
Y	$[0, s-1]$	$[s, 2s+r-2]$	$[2s+r-1, n-1]$
Z	$[0, s+r-2]$	$[s+r-1, 2s+r-2]$	$[2s+r-1, n+r-2]$

Encode the second input as $g_\varepsilon = g_L + \varepsilon t^s q + t^{2s+r-1} g_R$. Decode by dividing the middle output block by ε; leave the outer output blocks unchanged.

These are diagonal linear maps on the second input and the output. The first input f is unchanged.

Lemma 2.1(3), appendix "The three-sector splitting". All intervals include both endpoints; the Y range is $0, \ldots, n-1$ and the Z range is $0, \ldots, n+r-2$.

Three-sector proof: only matched blocks survive

A nonzero term has $c = a + b$ with $0 \le a \le r - 1$.

$$b \le s - 1 \implies c \le s + r - 2 \quad \text{(left } Y \text{ implies left } Z\text{)},$$
$$b \ge 2s + r - 1 \implies c \ge 2s + r - 1 \quad \text{(right } Y \text{ implies right } Z\text{)}.$$

The multipliers on the possible blocks are therefore

$Y \backslash Z$	Left	Middle	Right
Left	1	0	0
Middle	ε	1	ε
Right	0	0	1

No supported term has a negative power of ε. As $\varepsilon \to 0$, only the three matched blocks remain.

$C_{XY}(r, n) \trianglerighteq T_1 + T_2 + T_3$, where T_1, T_2 are the left and right matched blocks, and T_3 is the middle matched block.

Zero means no supported terms. An ε-block vanishes (and may be empty); the middle–middle factor is $\varepsilon\varepsilon^{-1} = 1$.

Three-sector proof: identify the outer blocks

The left block has $0 \le b < s$. Every $a + b$ lies in the left Z interval, so

$$T_1 = \sum_{a=0}^{r-1} \sum_{v=0}^{s-1} x_a y_v z_{a+v} = C_{XY}(r, s).$$

Put $A = 2s + r - 1$. In the right block, write $b = A + v$, with $0 \le v < s$:

$$T_2 = \sum_{a=0}^{r-1} \sum_{v=0}^{s-1} x_a y_{A+v} z_{A+a+v}.$$

Relabel $y'_v = y_{A+v}$ and $z'_d = z_{A+d}$. Then

$$T_2 = \sum_{a=0}^{r-1} \sum_{v=0}^{s-1} x_a y'_v z'_{a+v} \cong C_{XY}(r, s).$$

Both outer tensors are copies of $C_{XY}(r, s)$. The first-leg basis is the same in both; their Y, Z coordinates are disjoint.

The displayed index changes are bijections on the corresponding coordinate intervals. The full ranges follow from the partition and $0 \le a < r$, $0 \le v < s$.

Three-sector proof: identify the middle block

Write $c = s + r - 1 + v$ and $a = r - 1 - u$, with $0 \le v < s$, $0 \le u < r$. Then $b = c - a = s + u + v \in [s, 2s + r - 2]$. These choices cover exactly the middle block:

$$T_3 = \sum_{u=0}^{r-1} \sum_{v=0}^{s-1} x_{r-1-u} y_{s+u+v} z_{s+r-1+v}.$$

Put $q_d = y_{s+d}$ and $h_v = z_{s+r-1+v}$. Reading X, Z as inputs, define their polynomials by

$$f^{\leftarrow}(t) = \sum_{u=0}^{r-1} x_{r-1-u} t^u, \qquad h(t) = \sum_{v=0}^{s-1} h_v t^v.$$

The coefficient of the output label q_d is

$$\sum_{u+v=d} x_{r-1-u} h_v = [t^d](f^{\leftarrow} h).$$

Thus $(f, h) \mapsto f^{\leftarrow} h$, giving $T_3 \cong C_{XZ}(r, s)$ by first-leg reversal.

Reversal identifies this sector individually; the degeneration keeps the common X coordinates. This proves Lemma 2.1(3); permuting the legs gives every orientation.

Backup: size bounds and input interchange

Size and monotonicity

For all $r, s > 0$,

$$\min\{r, s\} \le g_{ij}(r, s) \le r + s.$$

The profile is nondecreasing in each length.

Interchanging inputs gives first-coordinate concavity

$$g_{ij}(r, s) = g_{ji}(s, r).$$

For fixed $s > 0$ and $p_j > 0$, the right-hand profile is powered-concave in its second argument. Thus

$$r \longmapsto g_{ij}(r, s)^{1/p_j} \quad \text{is concave.}$$

Lemma 4.1 and Appendix F. The power for varying the first length is $1/p_j$, obtained by exchanging the two polynomial inputs.

Backup: why degenerations preserve character bounds

For a degeneration $T \trianglerighteq S$, interpolation (slide 20) gives

$$(NL+1)\,T^{\otimes N} \geq S^{\otimes N}.$$

Apply character additivity, multiplicativity, and restriction monotonicity:

$$\lambda(S)^N \leq (NL+1)\lambda(T)^N.$$

Take N-th roots:

$$\lambda(S) \leq \underbrace{(NL+1)^{1/N}}_{\longrightarrow 1}\lambda(T).$$

Let $N \to \infty$: $T \trianglerighteq S$ implies $\lambda(S) \leq \lambda(T)$.

Lemma 3.1 and Appendix D; [5, Theorem 8]. Here L is the degree of the transformed polynomial tensor in the interpolation construction.

Backup: zero character parameters

The common-slope lemma supplies the offsets in row i when $p_i > 0$. When $p_i = 0$, define

$$c_{ij} = c_{ik} = 0.$$

Every row still has nonnegative offsets with sum at most one. A pair involving $p_i = 0$ satisfies, for all $r, s > 0$,

$$\frac{p_i s}{s + c_{ij} r} + \frac{p_j r}{r + c_{ji} s} = \frac{p_j r}{r + c_{ji} s} \leq p_j \leq 1.$$

Pairs with two positive parameters use the finite-increment proof.

The weighted reciprocal bound applies to every character, including zero parameters.

Propositions 5.2–5.3. No reciprocal power or common slope is assigned to a zero-parameter row.

Backup: formalization and provenance

Proof routes and scope.

Pinned finite splitting certificates and local-polynomial replication differ from the manuscript's exact constructions. The formalization also retains the angular route to weighted bounds and checks field descent; the slides follow the note's complex-field presentation. Applications are not formalized.

The accompanying note, its proofs, analysis and writeup were obtained fully through OpenAI Codex (GPT-6.1). The author has read and analyzed the note in full, takes full responsibility for its contents, and takes no credit for this work. These slides are a preliminary presentation of the note.

Repository: `lean/README.md` and `lean/PROOF-MAP.md`. Source theorem input: [2, 3].

References: proof ingredients

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